Lecture Material #2

Information Security

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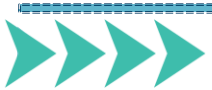
Faculty Profile

Hashing



Hashing

Hash tables are used for keeping values with a key

Just imagine a locker. You can only open them if you have the keys.

Hashing

Assume you have a hash table named **Users** wherein username is used as key and the value is the name. It will be like this:

1st record in Hash Table
Key: jsmith
Value: John Smith

2nd record in Hash Table
Key: jdoe
Value: Jane Doe

Hashing

Hashing is a **fundamental building block of cryptography**.

It is used for **security, integrity, and authentication**, not for hiding data.

Important distinction

- **Encryption** → reversible (decrypt to get original data)
- **Hashing** → **irreversible** (cannot get original data back)

Applications of Hashing

1 Password Storage

Instead of storing passwords:

```
password123
```

```
hash(password123)
```

```
entered password → hash → compare
```

✔ Even if database is leaked, original passwords are safe.

Applications of Hashing

2 Data Integrity

Hash ensures data is not modified.

Example:

- File hash before sending
- File hash after receiving
- If hashes match → file unchanged

Applications of Hashing

3 Digital Signatures

In public-key cryptography:

- Message → hash
- Hash is signed using private key
- Receiver verifies hash using public key

✔ Ensures **authenticity**, **integrity**, and **non-repudiation**

Applications of Hashing

4 Message Authentication Codes (MAC, HMAC)

Used to verify:

- Message integrity
- Sender authenticity

Example:

```bash
HMAC(hash function, secret key, message)
```

Applications of Hashing

5 Blockchain & Cryptocurrencies

Hashing is the backbone of:

- Block linking
- Proof of Work
- Tamper resistance

Properties of Hash Function

Property 1: Deterministic

No matter how many times you parse through a particular input through a hash function you will always get the same result.

Property 2: Quick Computation

The hash function should be capable of returning the hash of an input quickly. If the process isn't fast enough then the system simply won't be efficient.

Properties of Hash Function

Property 3: Pre-Image Resistance

What pre-image resistance states is that given H(A) it is infeasible to determine A, where A is the input and H(A) is the output hash.

Property 4: Small Changes In Input Changes the Hash

Even if you make a small change in your input, the changes that will be reflected in the hash will be huge.

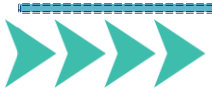
Properties of Hash Function

Property 5: Collision Resistant

Given two different inputs A and B where H(A) and H(B) are their respective hashes, it is infeasible for H(A) to be equal to H(B).

Property 6: Puzzle Friendly

It should be difficult to select an input that provides a pre-defined output. Thus, the input should be selected from a distribution that's as wide as possible.



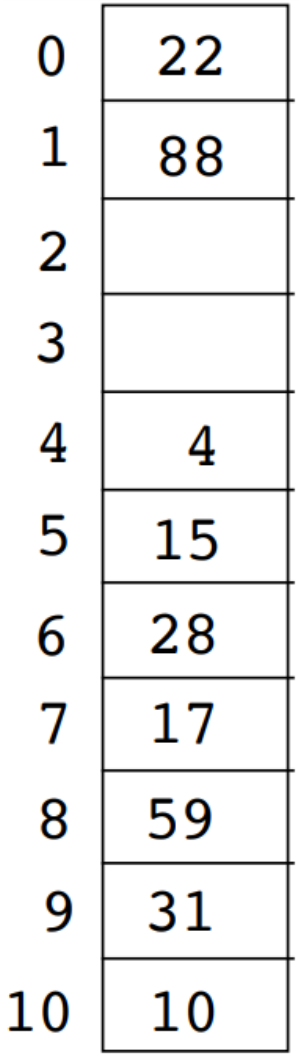
Hashing

Consider inserting the keys

10, 22, 31, 4, 15, 28, 17, 88, and 59

into a hash table of length m=11 using open addressing with the primary hash function h(k) = k mod m. Illustrate the result of inserting these keys using collision avoidance through linear probing. What is the resultant hash table?

Hashing

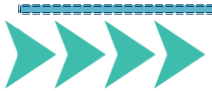


Hashing (Linear Probing)

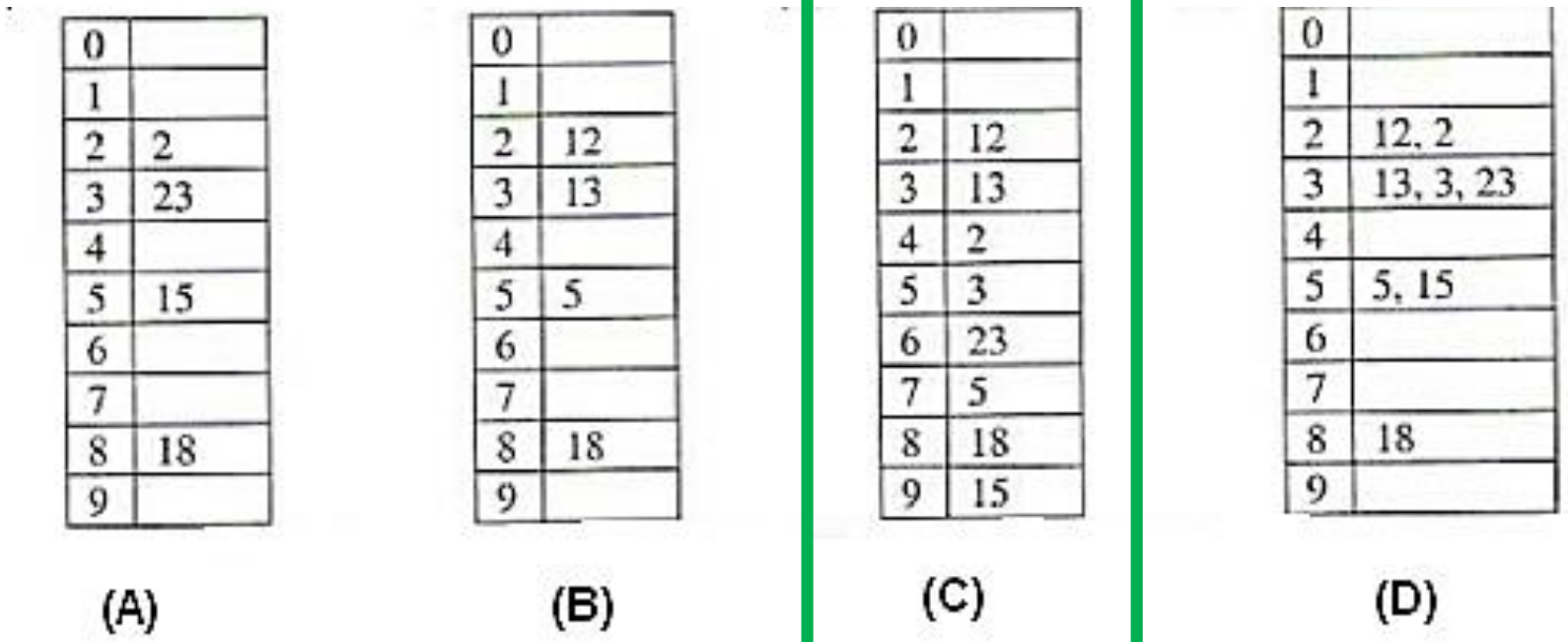
Consider inserting the keys

12, 18, 13, 2, 3, 23, 5 and 15

into a hash table of length m=10 using open addressing with the primary hash function h(k) = k mod m. Illustrate the result of inserting these keys using collision avoidance through linear probing.



Hashing (Linear Probing)



0	
1	
2	2
3	23
4	
5	15
6	
7	
8	18
9	

(A)

0	
1	
2	12
3	13
4	
5	5
6	
7	
8	18
9	

(B)

0	
1	
2	12
3	13
4	2
5	3
6	23
7	5
8	18
9	15

(C)

0	
1	
2	12, 2
3	13, 3, 23
4	
5	5, 15
6	
7	
8	18
9	

(D)

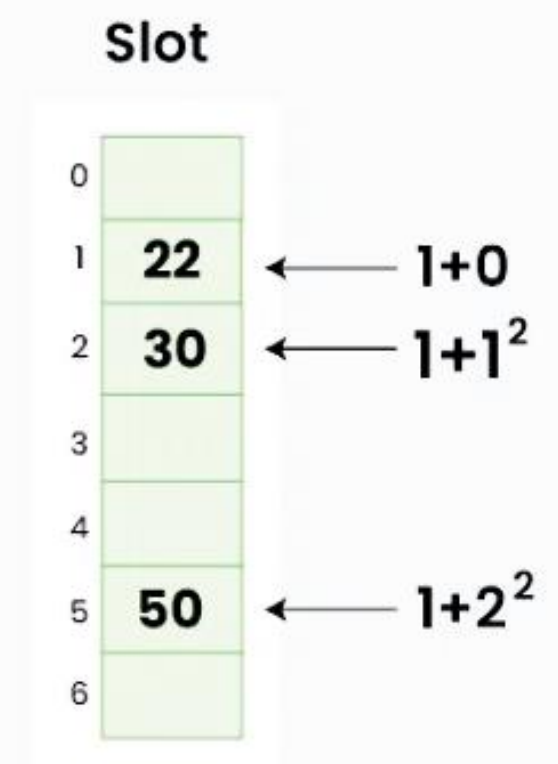
Hashing (Quadratic Probing)

let hash(x) be the slot index computed using hash function.

*If slot hash(x) % S is full, then we try (hash(x) + 1*1) % S*
*If (hash(x) + 1*1) % S is also full, then we try (hash(x) + 2*2) % S*
*If (hash(x) + 2*2) % S is also full, then we try (hash(x) + 3*3) % S*

Hashing (Quadratic Probing)

Example: Let us consider table Size = 7, hash function as Hash(x) = x % 7 and collision resolution strategy to be f(i) = i^2 . Insert = 22, 30, and 50.



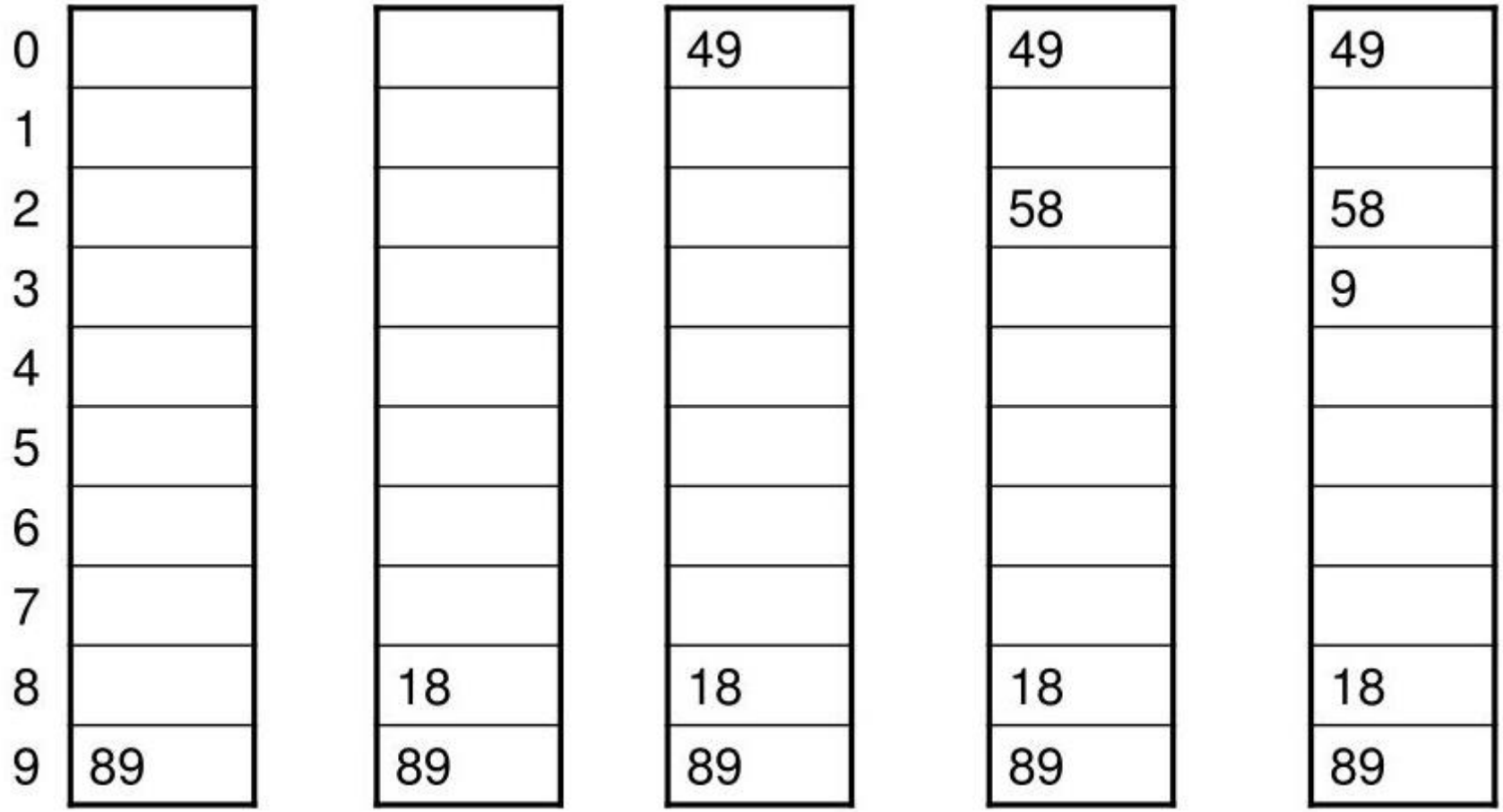
Hashing (Quadratic Probing)

Practice Problem

Insert: 89, 18, 49, 58, 9 to table size=10, hash function is: %tablesize

Hashing (Quadratic Probing)

Insert: 89, 18, 49, 58, 9 to table size=10, hash function is: %tablesize



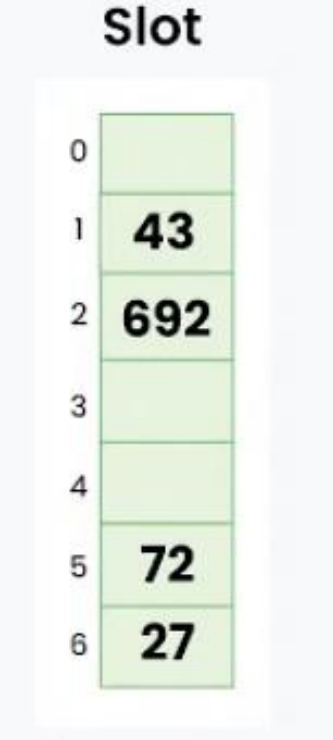
Hashing (Double Hashing)

let hash(x) be the slot index computed using hash function.

*If slot hash(x) % S is full, then we try (hash(x) + 1*hash2(x)) % S*
*If (hash(x) + 1*hash2(x)) % S is also full, then we try (hash(x) + 2*hash2(x)) % S*
*If (hash(x) + 2*hash2(x)) % S is also full, then we try (hash(x) + 3*hash2(x)) % S*

Hashing (Double Hashing)

Example: Insert the keys 27, 43, 692, 72 into the Hash Table of size 7. where first hash-function is h1(k) = k mod 7 and second hash-function is h2(k) = 1 + (k mod 5)



The next key is **72** which is mapped to **slot 2** **(72 % 7 = 2)**, but location **2** is already occupied. Using double hashing,

hnew = [h1(72) + i * (h2(72)] % 7
= [2 + 1 * (1 + 72 % 5)] % 7
= 5 % 7
= 5,

Now, as **5** is an empty slot, so we can insert **72** into **5th slot**.

Hashing (Double Hashing)

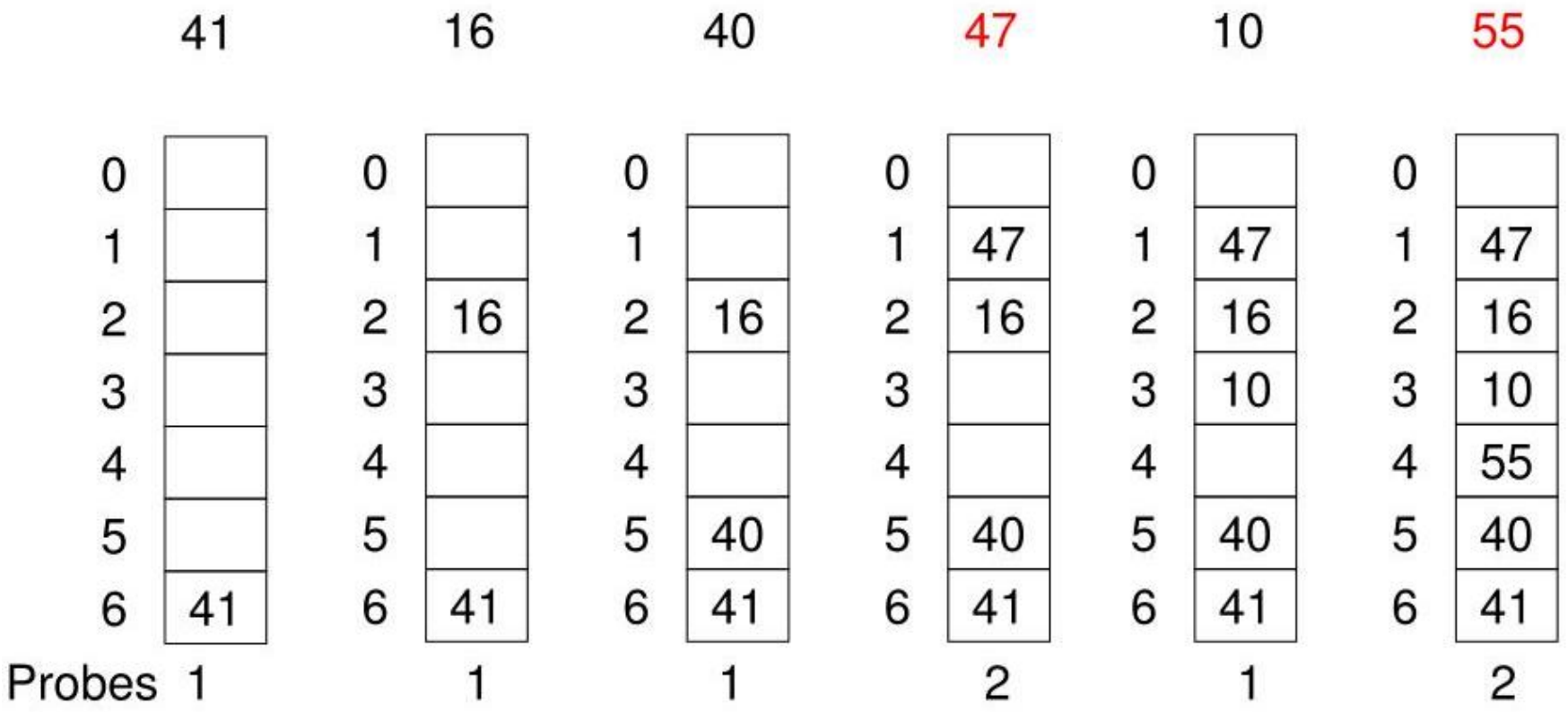
Practice Problem

$h(x) = x \% 7$ and $g(x) = 5 - (x \% 5)$

41 16 40 47 10 55

Hashing (Double Hashing)

$h(x) = x \% 7$ and $g(x) = 5 - (x \% 5)$




Thank You!